Answers on
Booklet - 12

Taken from OCR & AQA Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a) sum of current(s) into a point / junction = sum of current(s) out (from the point / junction) B2
(-1 for omission of 'point' or 'sum' in the statement of the law)
(Algebraic sum of current(s) at a point = 0 scores 2/2)

(b)(i) Thermistor B1

(b)(ii) $I_1 = 51$ (mA) B1
$I_2 = 9$ (mA) B1
$I_3 = 29$ (mA) B1

2.

(a) (i) any valid example - e.g. LIGHT, MICROWAVES (any em waves) ------ B1 [1]
(allow "water" /"sea" but reject 'slinky' unless explained/shown)

(ii) any valid example: e.g. SOUND ----------B1 [1]
(allow 'pressure wave'; reject "water" and 'slinky' unless explained/shown)

(b)(i) *transverse* = vibrations perpendicular to wave (direction) (WTTE) ----------B1 [1]
(allow "motion is perpendicular to wave", reject vague answers: e.g "vibrate up+down")

(ii) *longitudinal* = vibrations parallel to wave direction (WTTE) ---------- - B1 [1]
(allow "motion is perpendicular to wave" reject vague answers: e.g "vibrate back and for)

Wave phenomenon	Transverse waves	Longitudinal waves	
REFLECTION	YES	YES	
REFRACTION	**YES**	**YES**	B1
DIFFRACTION	**YES**	**YES**	B1 [3]
POLARISATION	**YES**	**NO**	B1

3.

a		reference to a transverse wave or to vibrations in plane normal to the direction of (energy) propagation	B1	can be answered with suitable diagram(s)
		oscillations/vibrations in one direction only/confined to single plane (containing the direction of propagation)	B1	NOT the wave oscillating in one plane
b		set up apparatus, e.g. tray of water on table with lamp/light from window	B1	
		rotate the filter	B1	QWC mark essential for full marks
		rotation of filter changes the image intensity/brightness/AW	B1	**allow** from bright to zero or vice versa
		correct orientation for maximum and minimum intensities of image	B1	transmission axis parallel to water surface for maximum and perpendicular for minimum
		move head up or down to change angle of reflected light observed	B1	can hold head still and move lamp
		use of protractor to measure angles	B1	
		image/reflection becomes partially plane polarised/ image changes from bright to dim but does not disappear	B1	**max 3** from 6 marking points **+ QWC mark**
c		$I = I_0 \cos^2\theta$	B1	
		where I_0 is the maximum intensity (of the polarised beam)	B1	**allow** incident/original/initial for maximum
		when θ is zero maximum intensity transmitted/ image bright	B1	
		when θ is 90° minimum/zero intensity transmitted/image dim/vanished	B1	

4.

(a) $Q = It$ (Allow any subject) C1

$Q = 0.040 \times 5.0 \times 60 \times 60$ \ $Q = 0.040 \times 1.8 \times 10^4$

charge = 720 A1

($40 \times 5 = 200$ or $0.040 \times 5 = 0.02$ or $40 \times 1.8 \times 10^4 = 7.2 \times 10^5$ scores 1/2)

coulomb \ C \ As B1

(b) It is less because the average current is less \ area (under graph) is less \ current 'drops' after 3 hours. B1

5.

(a)(i) Coulomb / C (Allow Ampere second / As) B1

(ii) Voltmeter B1

(b)(i) P = VI (Allow other variant) C1

V = 36 / 3.0 C1

p.d. = 12 (V) A1

(ii) E = 36 x 600 C1

energy = 2.1(6) x 10^4 (J) ≈ 2.2 x 10^4 (J) A1

(iii) $\Delta Q = I\Delta t$ / Q = It (Allow other variant) C1

ΔQ = 3.0 x 600 C1

charge = 1.8 x 10^3 (C) A1

(iv) N = 1.8 x 10^3 / 1.6 x 10^{-19} (Possible ECF) C1

number = 1.1(3) x 10^{22} ≈ 1.1 x 10^{22} A1

6.

(a)	DIFFRACTION	B1	[1]
(b)	a constant phase difference/relation (WTTE) (allow "zero phase difference" and "in phase")	B1	[1]
(c)	constructive interference produces bright lines <u>AND</u> destructive for dark lines	B1	
	in phase for bright AND antiphase (allow 'out of phase') for dark (ALLOW diagrams showing crests/troughs meeting crests/troughs for this mark)	B1	
	path difference = whole number of wavelengths {allow nλ} for bright (ALLOW path difference = λ but NOT path difference = ZERO)	B1	
	path difference = odd number of half wavelengths {allow (n+1/2)λ} for dark (ALLOW path difference = ½λ)	B1	[4]
(d)	recall of $\lambda = ax/D$ (in any valid form)	C1	
	valid substitution: e.g. $x = (6.5\times10^{-7}\times1.5)/0.25\times10^{-3}$	C1	
	$x = \mathbf{3.9\times10^{-3}}$ m (3.9×10^{-6} scores 2 marks)	A1	[3]

7.

(a)	(i)	$= 590\times10^{-9}$ m ✓ (using $d\sin\theta = n\lambda$ gives) $\sin\theta = \frac{n\lambda}{d}$ or $= \frac{2\times590\times(10^{-9})}{1.67\times10^{-6}}$ ✓ = **0.707** or 7.07×10^{8} if nm used ✓ θ = 45.0° ✓ (accept 45°)	
	(ii)	(sin θ ≤ 1) gives $\frac{n\lambda}{d} \le 1$ or $n \le \frac{d}{\lambda}$ or $= \frac{1.67\times10^{-6}}{590\times10^{-9}}$ ✓ = 2.83 ✓ so 3rd order or higher order is not possible ✓ alternative solution: (substituting) $\boldsymbol{n = 3}$ (into $d\sin\theta = n\lambda$ gives) ✓ $\sin\theta\ (=\frac{n\lambda}{d} = \frac{3\times590\times10^{-9}}{1.67\times10^{-6}}) = 1.06$ ✓ gives 'error'/which is not possible ✓	**7**
(b)		(using $d\sin\theta = n\lambda$ gives) $2\lambda = 1.67\times10^{-6}\times\sin 42.1$ ✓ $\lambda\ (= 0.5\times1.67\times10^{-6}\times\sin 42.1) = 5.6(0)\times10^{-7}$ m (or 560 nm) ✓	**2**

8.

$R = \frac{R_1 R_2}{R_1 + R_2}$ / $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ / $R = \frac{180 \times 120}{180 + 120}$ / $\frac{1}{R} = \frac{1}{120} + \frac{1}{180}$ **C1**

R = 72 (Ω) (-1 for each error) **A1**

R_{XY} = 72 +100 = 172 (Ω) (Allow 170) (Possible ECF) **A1**

9.

(Sum of) e.m.f.s = sum / total of p.d.s / sum of voltages (in a loop) **B1**
(Do not allow equation unless the symbols are defined)
Energy is conserved **B1**

10.

(a) $R = R_1 + R_2$ / $R = 200 + 120$ / $R = 320$ C1

current $= \frac{8.0}{320}$ C1

current $= 2.5 \times 10^{-2}$ (A) A0

(b) $V = 25 \times 10^{-3} \times 120$ / $V = \frac{120}{120 + 200} \times 8.0$

$V = 3.0$ (V) (Possible ecf) B1

(c) p.d. across the 360 (Ω) resistor = p.d. across the 120 (Ω) resistor /
There is no current between **A** and **B** / in the voltmeter B1
(Allow '*A & B have same voltage*' - BOD)

The p.d. calculated across 360 Ω resistor is shown to be 3.0 V /
The ratio of the resistances of the resistors is shown to be the same. B1

11.

a	i	A diagram with three resistors in <u>series</u> total resistance = 44 (Ω)	B1 B1	There is one mark for the diagram and one mark for the resistance value – the marks are **independent** of each other The resistors values are not necessary on the diagram The resistors must be shown as oblongs
	ii	A diagram with three resistors in <u>parallel</u> Use of $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}(+\frac{1}{R_3})$ $R=1/(10^{-1}+12^{-1}+22^{-1})$ total resistance = 4.37 (Ω)	B1 C1 A1	There is one mark for the diagram and two marks for the resistance value – the marks are **independent** of each other The resistors values are not necessary on the diagram. The resistor symbol is oblong or consistent with (a)(i) **Allow** A bald 4.37 (Ω) or 4.4 (Ω) scores two marks for calculation Answer can be 2 sf or more **Special case**: 4 (Ω) scores 1 mark for the calculation part

12.

(a) Finite resistance at 0° C — B1
Resistance increases — B1

(b)(i) Any <u>four</u> from:

1. The resistance of the <u>thermistor</u> decreases (as temperature is increased) — B1
2. The <u>total</u> resistance (of circuit) decreases — B1
3. The voltmeter reading increases — B1
4. Explanation of 3. above in terms of 'sharing voltage' / $\frac{V_1}{V_2}=\frac{R_1}{R_2}$ / $V=\frac{R_2}{R_1+R_2}\times V_0$ — B1
5. The current increases / ammeter reading increases — B1
6. Explanation of current increase in terms of $I=\frac{V}{R_{total}}$ — B1

(Allow ecf for statements 3. and 5. if statement 1. is incorrect – maximum score of 2/4)

(b)(ii) $I=\frac{3.6}{1200}(=3.0\times10^{-3})$ / $\frac{V_1}{V_2}=\frac{R_1}{R_2}$ / $V=\frac{R_2}{R_1+R_2}\times V_0$ — C1

$R=\frac{1.4}{3.0\times10^{-3}}$ / $\frac{R}{1200}=\frac{1.4}{3.6}$ / $1.4=\frac{R}{R+1200}\times5.0$ — C1

R = 467 (Ω) ≈ 470 (Ω) — A1

(When 1.4 V and 3.6 V are interchanged, then R = 3.1 × 10^3 (Ω) can score 2/3)

(Calculation of total circuit resistance of 1.67 × 10^3 (Ω) can score 2/3)

(Use of $I=\frac{5.0}{1200}$ scores 0/3)

13.

(a)(i)	One tick only for JC^{-1}	B1
(ii)	Σ e.m.f. = 4.5 – 1.5 (= 3.0)	C1
	V = 3.0 - 0.2 = 2.8 (V) (Ignore sign)	A1
(b)(i)	$V = V_0 \times R_2 / R_1 + R_2$ | $R = V/I$ <u>and</u> $R = R_1 + R_2$	C1
	$V = 5.0 \times 4.7 / (2.4 + 4.7)$ | $I = 5.0 / 7100 = 7.04 \times 10^{-4}$ (A)	C1
	| $V = 7.04 \times 10^{-4} \times 4700$	
	$V = 3.31 \approx 3.3$ (V) | $V = 3.31 \approx 3.3$ (V)	A1
	(Answer of $1.69 \approx 1.7$ (V) scores 2/3)	
(ii)	Decreases (Do not allow the mark if there is a contradiction)	B1